JAMA | Original Investigation

Colorectal Cancer and Mortality Risk Among Older Adults With vs Without Adenoma on Prior Colonoscopy

Samir Gupta, MD, MSCS; Lin Liu, PhD; Joshua Demb, PhD; Jiyue Qin, MS; Ashley Earles, MPH; Mark Lamm, MPH; Xinyi Huang, MS; Shailja C. Shah, MD; Neetu Chawla, PhD; Yi Le, MS; Tamra Burns Loeb, PhD; Jason A. Dominitz, MD, MHS; Douglas J. Robertson, MD, MPH; Nancy L. Schoenborn, MD; Audrey H. Calderwood, MD, MS; Theodore R. Levin, MD; Benjamin J. Seligman, MD, PhD; Folasade P. May, MD, PhD

IMPORTANCE Colorectal cancer (CRC) risk among older adults with prior adenoma is uncertain.

OBJECTIVE To estimate cumulative CRC risks, non-CRC mortality, and all-cause mortality among adults 75 years of age or older with vs without adenoma at prior colonoscopy (in the latter of whom, guidelines recommend against repeat colonoscopy screening).

DESIGN, SETTING, AND PARTICIPANTS Retrospective cohort study of older adults who underwent colonoscopy between January 1, 2006, and December 31, 2019, and prior to 75 years of age within the US Department of Veterans Affairs.

EXPOSURES Colonoscopy with vs without adenoma prior to 75 years of age.

MAIN OUTCOMES AND MEASURES Estimated cumulative incidence of CRC, CRC death, non-CRC death, and all-cause mortality for individuals with vs without adenoma at prior colonoscopy (incidence of CRC and CRC death were compared using the Gray test). For those with adenoma, incidence of CRC and non-CRC death were stratified based on 5 Veterans Affairs Frailty Index categories of increasing all-cause mortality risk (nonfrail, ≤0.10; prefrail, 0.11-0.20; mild frailty, 0.21-0.30; moderate frailty, 0.31-0.40; and severe frailty, >0.40).

RESULTS Of 91 952 individuals (median age, 71 [IQR, 69-73] years at last colonoscopy; 98% male) who had undergone colonoscopy prior to 75 years of age, there were 25 538 (27.8%) with adenoma vs 66 414 (72.2%) without adenoma. At 10-year follow-up, the cumulative incidence of CRC was 1.1% (95% CI, 0.8%-1.3%) in those with adenoma vs 0.7% (95% CI, 0.5%-0.8%) in those without adenoma (Gray test $P < .001$). At 10-year follow-up, the cumulative incidence of CRC death was 0.5% (95% CI, 0.3%-0.7%) in those with adenoma vs 0.4% (95% CI, 0.3%-0.5%) in those without adenoma (Gray test $P = .005$). The cumulative incidence of non-CRC death ranged from 46.9% to 48.4% at 10 years. For those with adenoma, incidence of CRC was substantially exceeded by the cumulative incidence of non-CRC death at 10-year follow-up across all frailty levels (ranged from 34.2% among nonfrail individuals to 82.0% among severely frail individuals).

CONCLUSIONS AND RELEVANCE Adults 75 years of age or older with adenoma at prior colonoscopy were more likely to experience subsequent CRC and CRC death compared with those without adenoma, but cumulative risks were low and were far exceeded by competing risks for non-CRC death. Older adults may consider deprioritizing surveillance colonoscopy relative to other health concerns.

JAMA. doi:10.1001/jama.2026.3414
Published online April 9, 2026.

Editorial

Supplemental content

Author Affiliations: VA San Diego Healthcare System, San Diego, California (Gupta, Liu, Earles, Lamm, Shah); University of California, San Diego (Gupta, Liu, Demb, Qin, Shah, Le); Veterans Medical Research Foundation, San Diego, California (Huang); VA Greater Los Angeles Healthcare System, Los Angeles, California (Chawla, Seligman, May); University of California, Los Angeles (Loeb, Seligman, May); VA Puget Sound Health Care System, Seattle, Washington (Dominitz); VA White River Junction Health Care, White River Junction, Vermont (Robertson); Johns Hopkins University, Baltimore, Maryland (Schoenborn); Dartmouth Health, Lebanon, New Hampshire (Calderwood); Permanente Medical Group Inc, Pleasanton, California (Levin).

Corresponding Author: Samir Gupta, MD, MSCS, VA San Diego Healthcare System, 3350 La Jolla Village Dr, MC 111D, San Diego, CA 92161 (s1gupta@health.ucsd.edu).

Adults 75 years of age or older (hereafter referred to as older adults) and their clinicians are increasingly faced with challenging decisions regarding whether to continue cancer screening and surveillance. Cancer risk increases with age, and cancer is the second leading cause of mortality among older adults.[1] Simultaneously, older adults face increasing competing risks for noncancer mortality, particularly with higher levels of frailty. In addition, for those who have previously received screening or surveillance, the incremental benefit of additional testing is unclear. These challenges are especially relevant to colorectal cancer (CRC) surveillance after polypectomy. In the US, nearly 75% of adults 65 years of age or older have had a colonoscopy.[2] The mean adenoma detection rate is 38% and may be as high as 60%.[3] After adenoma removal, usual practice is to perform surveillance colonoscopy for early detection of CRC and for its prevention through identification and removal of polyps.[4]

For older adults, the benefits of continued surveillance are uncertain, and current guidelines do not provide clear age or frailty-based recommendations for deimplementation of surveillance. This evidence gap is of high clinical relevance because colonoscopy and sedation risks increase with age and frailty and because colonoscopy is a finite, costly resource.[5,6] Yet, CRC surveillance is commonly performed in older adults, even among those with limited life expectancy.[7] Knowledge of CRC incidence and mortality risk for older adults with vs without adenomatous polyps at prior colonoscopy, considered in the context of non-CRC and all-cause mortality, may help inform decision-making surrounding surveillance colonoscopy. Our aim was to use a large national sample of adults 75 years of age or older with vs without adenoma at prior colonoscopy to examine risk for CRC, non-CRC death, and all-cause mortality.

Key Points

Question What are the risks for colorectal cancer (CRC) and all-cause mortality among adults 75 years of age or older with or without adenoma at prior colonoscopy, the latter of whom guidelines recommend against repeat screening?

Findings Among US veterans who underwent colonoscopy prior to 75 years of age, subsequent 10-year incidence of non-CRC death ranged from 46.9% to 48.4%, which far exceeded cumulative incidence of CRC (1.1%) and CRC death (0.5%) in those with prior adenoma vs 0.7% and 0.4%, respectively, in those without adenoma.

Meaning Among adults 75 years of age or older, the risk of death from causes other than CRC greatly exceeded the risk of diagnosis or CRC-specific death, regardless of prior adenoma status.

Methods

We conducted a retrospective cohort study among US veterans who underwent colonoscopy between January 1, 2006, and December 31, 2019, within the Department of Veterans Affairs (VA), which is one of the largest health care systems in the US, using usual care electronic health record data. In reporting the results, we followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline. The study was approved by the VA Central, VA San Diego, and VA Greater Los Angeles institutional review boards with a waiver of informed consent and is part of the Surveillance Colonoscopy in Older Adults initiative.

We included individuals 75 years of age or older who underwent a colonoscopy at or after 65 years of age and had either detection of 1 or more adenomas or no adenoma. Adenoma detection was ascertained based on microscopic diagnoses. Although detection did not include verification of complete polyp excision, complete removal of polyps was standard practice through the observation period. The colonoscopy performed closest in time prior to turning 75 years of age was used for ascertaining adenoma status. Individuals with incomplete extent of colonoscopy examination, inadequate bowel preparation, missing documentation of extent or bowel preparation, CRC or inflammatory bowel disease diagnosis prior to 75 years of age, or prior history of sessile serrated lesions without a concurrent adenoma were excluded. Data on extent of examination, quality of bowel preparation, and colonoscopy findings, including number, size, and histology of polyps reported, were generated by previously validated natural language processing algorithms.[8]

Outcomes

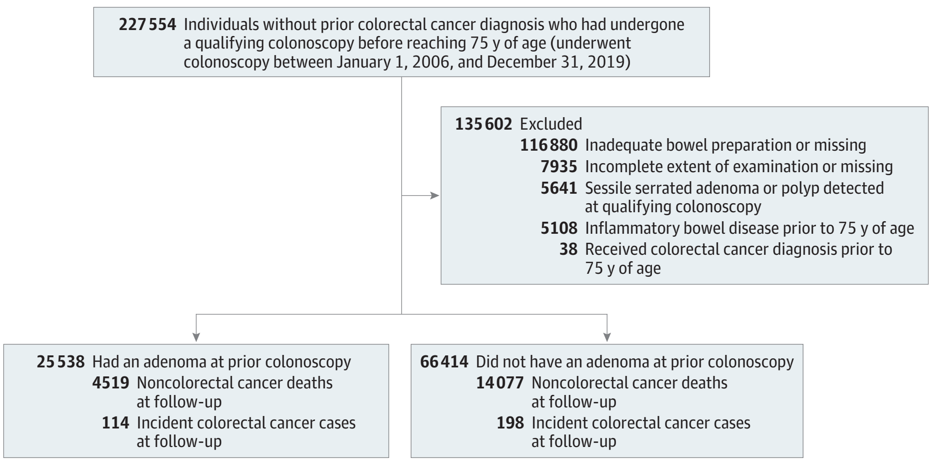
The primary outcomes of interest included incident CRC (ascertained via VA oncology domain cancer registry data), CRC death (ascertained via National Death Index–linked primary cause of death data), and non-CRC death and all-cause mortality (ascertained via National Death Index data). The primary predictors were exposure to colonoscopy prior to 75 years of age with or without adenoma detection. The rationale for including individuals without adenoma at colonoscopy is based on the US Preventive Services Task Force grade C recommendation for selective CRC screening for adults older than 75 years, noting that people exposed to prior screening such as colonoscopy with normal findings are likely to benefit minimally.[9]

We postulated finding similar CRC risk for older adults with adenoma at prior colonoscopy as for those without adenoma would support analogous recommendations regarding surveillance for older adults. Follow-up began at an individual's 75th birthday and continued until the earliest (1) incident CRC, death, or December 31, 2019, for analyses focused on incidence of CRC and (2) CRC death, non-CRC death, or December 31, 2019, for the death analyses. For 51 patients with National Death Index–linked CRC death but without a documented diagnosis date in the oncology domain or in VA-linked Medicare claims data, the CRC death date was used as the diagnosis date for incident CRC. Clinical characteristics collected as part of usual medical care such as age, sex, and race and ethnicity (based on self-report to fixed categories) were extracted from electronic health record data for the entire cohort.

Statistical Analysis

To evaluate incident CRC risk among those with adenoma at prior colonoscopy, we estimated cumulative incidence of CRC; non-CRC death was assessed as a competing risk.[10-13] For descriptive comparison, we also estimated the cumulative incidence of CRC for those without adenoma and the cumulative

Figure 1. Flow Diagram of Participants Through Study



incidence of non-CRC death for those with vs without adenoma. The between-group differences in incident CRC were assessed using the Gray test.[11] The between-group differences in all-cause mortality were assessed using the log-rank test. A 2-sided P value <.05 was considered statistically significant.

In a secondary analysis, the cumulative incidence of CRC, CRC death, and non-CRC death was summarized at 5- and 10-year follow-up with associated 95% CIs for those with vs without adenoma. For visual comparisons, we generated cumulative incidence plots for CRC, CRC death, and all-cause mortality for those with and without adenoma.

In secondary analyses for those with adenoma, we estimated and plotted cumulative incidence of CRC and non-CRC death according to level of frailty using the Veterans Affairs Frailty Index.[14,15] The index is a previously validated measure that uses VA and VA-linked Medicare administrative claims data and categorizes frailty into categories associated with increasing risk for all-cause mortality (nonfrail, prefrail, mild frailty, moderate frailty, and severe frailty). The VA Frailty Index has been shown to have construct validity compared with direct clinical measures of frailty. Among those with adenoma, we evaluated cumulative incidence of CRC according to adenoma risk (advanced vs nonadvanced). *Advanced adenoma* was defined as 10 mm or larger in size, any size with villous or tubulovillous features, and any size with high-grade dysplasia. *Nonadvanced adenoma* was defined as any adenoma without the advanced adenoma features.

Time since the most recent colonoscopy prior to 75 years of age was considered a factor that might influence CRC risk, therefore, the incidence analyses for CRC in those with vs without adenoma were repeated and stratified by 3 time frames of exposure to colonoscopy (≤3 years, >3-5 years, >5-10 years) prior to reaching 75 years of age and when reaching 75 years.

In an exploratory analysis for those with adenoma, influence of exposure vs no exposure to colonoscopy after 75 years of age on incident CRC was evaluated by examining distribution of the number of follow-up colonoscopies after reaching 75 years among individuals who developed vs did not develop CRC using the Fisher exact test. In this analysis, the follow-up exposure was defined as a colonoscopy that occurred after 75 years of age in which no CRC was diagnosed within 6 months.

We also conducted 2 sensitivity analyses. Given the significant number of people excluded, the first sensitivity analysis included individuals irrespective of missing information regarding bowel preparation or extent of examination or individuals with known inadequate bowel preparation or examination extent. The second sensitivity analysis excluded 51 individuals with CRC as the cause of death who did not have a documented preceding CRC diagnosis date from the incident CRC analysis.

Results

We included 91 952 individuals 75 years of age or older with adenoma (n = 25 538; 27.8%) found on a colonoscopy obtained before they reached 75 years vs without adenoma (n = 66 414; 72.2%). The median age at the most recent colonoscopy was 71 years (IQR, 69-73 years) and 98% were male. Most study exclusions were due to inadequate quality or missing documentation for bowel preparation (**Figure 1**). There were nonclinically significant between-group differences in the demographic and clinical characteristics. Compared with individuals without adenoma, those with adenoma were older, there were fewer females, there were more Black and Hispanic individuals, and there were more individuals who were nonfrail (**Table 1**).

At 10-year follow-up, the cumulative incidence of CRC was 1.1% (95% CI, 0.8%-1.3%) among individuals with adenoma vs 0.7% (95% CI, 0.5%-0.8%) in those without adenoma (Gray test P < .001; **Figure 2**A and **Table 2**). At 10-year follow-up, the cumulative incidence of CRC death was 0.5% (95% CI, 0.3%-0.7%) among individuals with adenoma vs 0.4% (95% CI, 0.3%-0.5%) in those without adenoma (Gray test P = .005; Figure 2B and Table 2).

Table 1. Demographic Characteristics

Characteristic	Adenoma status at prior colonoscopy[a]	
	Adenoma (n = 25 538)	No adenoma (n = 66 414)
Age at index colonoscopy, median (IQR), y	72 (70-73)	71 (69-73)
Sex		
Female	359 (1.4)	1453 (2.2)
Male	25 179 (98.6)	64 961 (97.8)
Race and ethnicity[b]	(n = 24 024)	(n = 61 751)
American Indian	128 (0.5)	347 (0.6)
Asian or Pacific Islander	262 (1.1)	724 (1.2)
Hispanic	1664 (6.9)	3569 (5.8)
Multiracial or other[c]	484 (2.0)	1254 (2.0)
Non-Hispanic Black	3112 (13)	6714 (11)
Non-Hispanic White	18 374 (76)	49 143 (80)
Veterans Affairs Frailty Index[d]		
Nonfrail (≤0.10)	13 080 (51.2)	31 773 (47.8)
Prefrail (0.11-0.20)	5913 (23.2)	16 836 (25.4)
Mild frailty (0.21-0.30)	3443 (13.5)	9540 (14.4)
Moderate frailty (0.31-0.40)	1746 (6.8)	4671 (7.0)
Severe frailty (>0.40)	1356 (5.3)	3594 (5.4)
Charlson Comorbidity Index[e]		
0	9262 (36.3)	26 460 (39.8)
1	5903 (23.1)	15 460 (23.3)
2	4086 (16.0)	9906 (14.9)
≥3	6287 (24.6)	14 588 (22.0)
Aspirin use	13 191 (51.7)	32 738 (49.3)
Baseline risk category		
Nonadvanced adenoma	22 772 (89.2)	0
Advanced adenoma[f]	2766 (10.8)	0
Body mass index[g]	(n = 19 007)	(n = 48 764)
<18.5	199 (1.0)	431 (0.9)
18.5-<25	3499 (18)	9584 (20)
25-<30	7513 (40)	20 242 (42)
≥30	7796 (41)	18 507 (38)
Period between index colonoscopy and 75th birthday, median (IQR), y	4 (2-6)	3 (2-5)
Length of follow-up for colorectal cancer, y		
<5	19 598 (77)	44 859 (68)
≥5 and <10	5131 (20)	17 747 (27)
≥10	809 (3.2)	3808 (5.7)

[a] Data are expressed as No. (%) unless otherwise indicated.

[b] Based on self-report and assignment that occurred as part of usual care.

[c] Includes people with more than 1 race recorded or who reported another race.

[d] Represents the proportion of 31 predefined health deficits on a scale from 0 to 1.[14,15] Derived from claims data using the cumulative deficit method. Higher scores predict increased mortality independent of age and other clinical predictors.

[e] Weighted scores based on 17 medical conditions. Higher scores reflect higher comorbidity burden and are associated with higher risk for mortality.

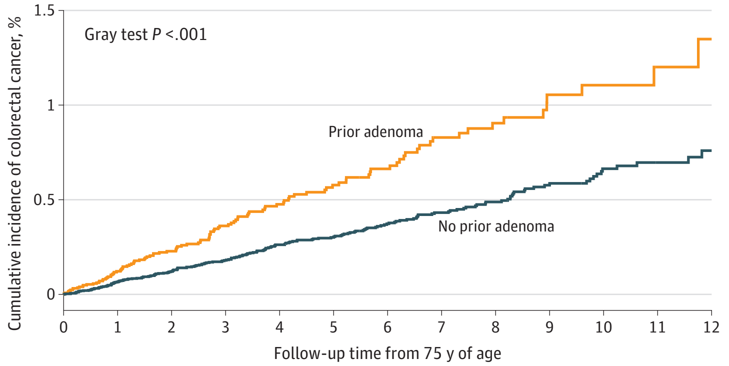
[f] Defined as 10 mm or larger in size, having villous or tubulovillous features, or having high-grade dysplasia.

[g] Calculated as weight in kilograms divided by height in meters squared.

The cumulative incidence of CRC and CRC death were substantially lower than the incidence of non-CRC death and all-cause mortality (eFigures 1-2 in Supplement 1 and Figure 2C). Among individuals with adenoma, the cumulative incidence of CRC at 5 years was 0.6% and non-CRC death was 22.4%; at 10 years, the cumulative incidence was 1.1% and 48.2%, respectively. Among individuals without adenoma, the cumulative incidence of CRC at 5 years was 0.3% and non-CRC death was 21.0%; at 10 years, the cumulative incidence was 0.7% and 46.9%, respectively (Table 2). Among individuals with adenoma, the cumulative incidence of CRC death at 5 years was 0.2% and non-CRC death was 22.4%; at 10 years, the cumulative incidence was 0.5% and 48.4%, respectively. Among individuals without adenoma, the cumulative incidence of CRC death at 5 years was 0.1% and non-CRC death was 21.0%; at 10 years, the cumulative incidence was 0.4% and 47.0%, respectively (Table 2).

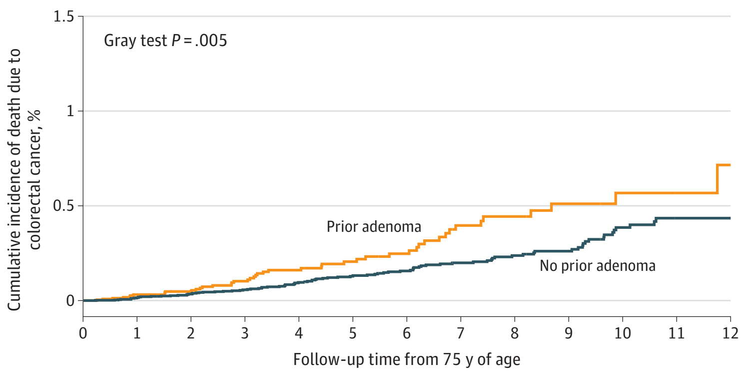
Figure 2. Survival Curves Showing the Cumulative Incidence of Colorectal Cancer, Death Due to Colorectal Cancer, and All-Cause Mortality

A Cumulative incidence of colorectal cancer



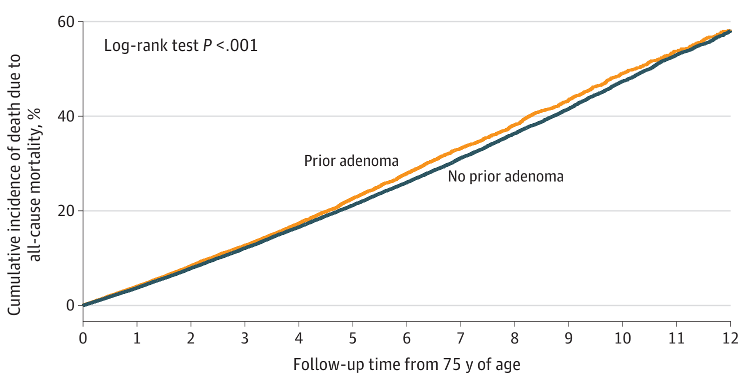
	0	1	2	3	4	5	6	7	8	9	10	11	12
Prior adenoma													
No. at risk	25538	19755	14810	10957	8108	5940	4354	3105	2126	1369	809	454	234
Colorectal cancer cases	0	28	48	67	80	89	95	104	107	111	112	113	114
No prior adenoma													
No. at risk	66414	54743	44213	34955	27616	21555	16552	12311	8843	6010	3808	2208	1043
Colorectal cancer cases	0	40	69	95	125	137	155	167	176	188	194	196	198

B Cumulative incidence of death due to colorectal cancer



	0	1	2	3	4	5	6	7	8	9	10	11	12
Prior adenoma													
No. at risk	25538	19776	14844	10994	8142	5972	4382	3128	2144	1382	819	461	237
Death due to colorectal cancer	0	7	10	18	25	29	32	40	42	44	45	45	46
No prior adenoma													
No. at risk	66414	54773	44251	35002	27674	21604	16604	12359	8883	6052	3839	2227	1053
Death due to colorectal cancer	0	9	19	29	43	54	60	69	75	78	89	92	92

C Cumulative incidence of all-cause mortality



	0	1	2	3	4	5	6	7	8	9	10	11	12
Prior adenoma													
No. at risk	25538	19776	14844	10994	8142	5972	4382	3128	2144	1382	819	461	237
All-cause mortality	0	931	1721	2346	2869	3329	3694	3978	4181	4338	4453	4515	4551
No prior adenoma													
No. at risk	66414	54773	44251	35002	27674	21604	16604	12359	8883	6052	3839	2227	1053
All-cause mortality	0	2286	4457	6317	7946	9336	10542	11580	12407	13041	13556	13889	14073

A, Cumulative incidence of colorectal cancer was higher in those with vs without adenoma. The median follow-up was 2.50 (IQR, 1.12-4.79) years for those with adenoma and was 3.23 (IQR, 1.46-5.99) years for those without adenoma. B, Cumulative incidence of death due colorectal cancer was higher in those with vs without adenoma. The median follow-up was 2.51 (IQR, 1.13-4.80) years for those with adenoma and was 3.23 (IQR, 1.46-6.00) years for those without adenoma. C, Cumulative incidence of death due to all-cause mortality was qualitatively similar in those with vs without adenoma. The median follow-up was 2.51 (IQR, 1.13-4.80) years for those with adenoma and was 3.23 (IQR, 1.46-6.00) years for those without adenoma.

At 10 years, the cumulative incidence of CRC was low among individuals with adenoma across all levels of frailty classification (1.7% in the nonfrail group and <1% in all other groups; eFigure 3A in Supplement 1). In contrast, the cumu-

Table 2. Secondary Colorectal Cancer (CRC) Outcomes

Outcome	Adenoma (n = 25 538)		No adenoma (n = 66 414)	
	No. of events	Cumulative incidence, % (95% CI)	No. of events	Cumulative incidence, % (95% CI)
5-y Incidence				
CRC[a]	89	0.6 (0.4-0.7)	137	0.3 (0.2-0.4)
Non-CRC death[b]				
Estimated cumulative incidence of non-CRC death vs CRC incidence[c]	3295	22.4 (21.6-23.1)	9272	21.0 (20.6-21.4)
Estimated cumulative incidence of non-CRC death vs CRC death	3300	22.4 (21.7-23.1)	9282	21.0 (20.6-21.4)
CRC death[d]	29	0.2 (0.1-0.3)	54	0.1 (0.1-0.2)
10-y Incidence				
CRC[a]	112	1.1 (0.8-1.3)	194	0.7 (0.5-0.8)
Non-CRC death[b]				
Estimated cumulative incidence of non-CRC death vs CRC incidence[c]	4395	48.2 (46.7-49.8)	13 445	46.9 (46.1-47.7)
Estimated cumulative incidence of non-CRC death vs CRC death	4408	48.4 (46.9-50.0)	13 467	47.0 (46.2-47.7)
CRC death[d]	45	0.5 (0.3-0.7)	89	0.4 (0.3-0.5)

[a] Defined by diagnosis within cancer registry data or having an event of CRC death recorded within National Death Index data. A person with incident CRC may have a non-CRC death.

[b] Defined by having primary cause of death other than CRC within the National Death Index data. Separate estimates are provided for competing risk for non-CRC death vs incident and fatal CRC because the outcomes for these analyses are different, resulting in small differences in the number of individuals at risk.

[c] A person with incident CRC may have had a non-CRC death.

[d] Primary cause of death within the National Death Index data.

lative incidence of non-CRC death at 10-year follow-up substantially exceeded the incidence of CRC across all levels of frailty (from 34.2% in the nonfrail group to 82.0% in the severe frailty group; eFigure 3B in Supplement 1).

Among individuals with adenoma, we did not find significant variation in the cumulative risk for incident CRC according to adenoma risk (eFigure 4 in Supplement 1). Furthermore, for individuals with adenoma, we found no significant variation in the frequency distribution of exposure to colonoscopy after 75 years of age (0, 1, 2, or ≥3 follow-up colonoscopies; eTable 1 in Supplement 1).

The results were qualitatively similar in the analyses (1) focused on individuals missing documentation on bowel preparation or on the extent of the examination or individuals who had inadequate bowel preparation or extent of the examination and (2) excluding the 51 individuals with CRC death and who were missing an incident CRC diagnosis date (results not shown).

The variation in years between the most recent colonoscopy and turning 75 years of age ranged from 1 to 10 years (eFigure 5 in Supplement 1) and the cumulative incidence of CRC remained low across these time intervals (eTable 2 in Supplement 1). Among the individuals who underwent colonoscopy 3 years or less before reaching 75 years of age, the 5-year cumulative incidence of CRC was 0.6% (95% CI, 0.5%-0.8%) in those with adenoma vs 0.3% (95% CI, 0.2%-0.4%) in those without adenoma. Among the individuals who underwent colonoscopy 3 years or more to 5 years before reaching 75 years of age, the 5-year cumulative incidence of CRC was 0.4% (95% CI, 0.2%-0.7%) in those with adenoma vs 0.3% (95% CI, 0.2%-0.4%) in those without adenoma. Among the individuals who underwent colonoscopy more than 5 years to 10 years before reaching 75 years, the 5-year cumulative incidence of CRC was 0.6% (95% CI, 0.3%-1.0%) in those with adenoma vs 0.3% (95% CI, 0.2%-0.4%) in those without adenoma.

Discussion

This study found that among adults in a large cohort who remained free from cancer to 75 years of age, those with adenoma found on a prior colonoscopy had a higher risk for subsequent incident CRC and CRC death vs those without adenoma on a prior colonoscopy. However, the absolute cumulative rates of incident CRC and CRC death were low for both groups of individuals.

Importantly, rates of incident CRC and CRC death were markedly exceeded by the competing risk of non-CRC death. In an exploratory analysis of individuals with prior adenoma, we did not find significant variation in frequency of exposure to follow-up colonoscopy after 75 years of age among those with incident CRC vs those without incident CRC. These findings raise major questions about the clinical relevance of surveillance colonoscopy in older adults with prior adenomas after reaching 75 years.

Longitudinal cohort studies have shown individuals with prior adenomas (particularly those with advanced adenomas) have higher risk for incident CRC and CRC death compared with those without adenoma at prior colonoscopy.[16-20] However, these longitudinal studies included few older adults, with a median or mean age ranging from 56 to 64 years, limiting generalizability.[16-19] A UK study reported cumulative incidence of CRC to be low at 393 (95% CI, 307-502) per 100 000 person-years of follow-up for adults 75 years of age or older after removal of intermediate risk adenomas (defined as 1-2 adenomas that were ≥1 cm in size or 3-4 adenoma <1 cm in size).[21]

A meta-analysis[22] of individuals with prior polyp removal reported 2.1% of adults aged 70 years or older compared with 1.4% of adults aged 50 to 70 years were diagnosed with CRC at time of follow-up surveillance colonoscopy. Across 9740 surveillance colonoscopy procedures performed among 9601 adults aged 70 to 85 years with a history of prior adenoma treated at a community-based regional health system, the proportion with CRC detected was 0.3% regardless of whether the interval between prior adenoma diagnosis and repeat colonoscopy was more than 1 year or more than 5 years.[23] Similarly, among 9831 adults aged 65 years or older who underwent colonoscopy for polyp surveillance, only 0.2% had CRC.[7]

Our findings confirm and extend prior work by offering longitudinal estimates of CRC risk by taking frailty into account, and, most importantly, by comparing these estimates with competing risks for non-CRC death. We postulate that accounting for frailty and competing risk for non-CRC death is critically relevant to clinical practice because clinicians and patients engaged in decision-making need to put CRC risk in context of a patient's personal health status and associated risk for non-CRC death. Our estimates of cumulative incidence of CRC are similar to the aforementioned 2 most recent cross-sectional studies of cancer rates after polypectomy.[7,23] Taken together with our comparisons of CRC risk with competing risk for non-CRC death, available data confirm that the likelihood of CRC among older adults with prior adenoma is very low.

These observations among older adults after polyp removal are consistent with very low rates of incident CRC relative to competing risk for non-CRC death reported among older adults who have had a normal fecal occult blood test[24] or normal colonoscopy.[25] Furthermore, the observed CRC risk among older adults with prior polypectomy may be too low to justify surveillance colonoscopy, particularly in light of well-established evidence that risks associated with colonoscopy increase with age and frailty.[26-29] Some studies have reported that 3.8% to 6.8% of older adults require an emergency department visit or hospitalization within 30 days of colonoscopy.[5,6]

In addition, the data from the current study show that the risk for non-CRC death far exceeds the risk for CRC, even across a range of levels of frailty. Notably, our exploratory analysis among older adults with prior adenoma did not find variation in frequency of exposure to follow-up colonoscopy among those with vs without incident CRC. Collectively, our results also highlight the importance of an ongoing clinical trial comparing deescalation of surveillance with a strategy of annual fecal immunochemical testing vs usual care surveillance colonoscopy among older adults with prior history of polypectomy.[30] Strengths of the study include the longitudinal cohort study design, large national sample from geographically diverse health centers, and our provision of incidence of CRC and mortality data in context of competing risk for non-CRC death and rates of all-cause mortality.

Limitations

Some limitations may be considered in interpreting our results. First, we studied a US veteran population that was almost entirely male. Studies are needed with higher representation of females to confirm consistency of results, particularly given that they have lower cumulative lifetime risk for CRC compared with males, yet also experience longer life expectancy under which to experience this risk and potentially benefit from surveillance.

Second, our exploratory analysis found no significant variation in the frequency of follow-up colonoscopy among individuals with prior adenoma with vs those without incident CRC and could not disaggregate risk reduction based on whether the indication for the follow-up colonoscopy was asymptomatic surveillance vs a diagnostic evaluation. A more detailed assessment of the effects of surveillance colonoscopy on CRC risk leveraging more granular, manually abstracted colonoscopy report data are the subject of a future study initiative.

Conclusions

Adults 75 years of age or older with adenoma at prior colonoscopy were more likely to experience subsequent CRC and CRC death compared with those without adenoma, but cumulative risks were low and were far exceeded by competing risks for non-CRC death. Older adults may consider deprioritizing surveillance colonoscopy relative to other health concerns.

ARTICLE INFORMATION

Accepted for Publication: March 5, 2026.

Published Online: April 9, 2026. doi:10.1001/jama.2026.3414

Author Contributions: Dr Gupta had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Gupta, Liu, Demb, Lamm, Shah, Chawla, Schoenborn, Levin, May.
Acquisition, analysis, or interpretation of data: Gupta, Liu, Demb, Qin, Earles, Huang, Le, Loeb, Dominitz, Robertson, Schoenborn, Calderwood, Seligman, May.
Drafting of the manuscript: Gupta, Liu, Demb, Lamm, Huang, Le, May.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: Gupta, Liu, Demb, Qin, Huang, Le.
Obtained funding: Gupta, Liu, May.
Administrative, technical, or material support: Gupta, Earles, Lamm, May.
Supervision: Gupta, Liu, May.

Conflict of Interest Disclosures: Dr Gupta reported receiving personal fees from Guardant Health, Geneoscopy, Universal Diagnostics, and Freenome. Dr Robertson reported receiving personal fees from Freenome and Amadix. Dr Levin reported receiving personal fees from Freenome, Geneoscopy, and Universal Diagnostics and serving in an advisory role with Navatar. Dr May reported receiving personal fees from Exact Sciences, Natera, Medtronic, and Geneoscopy. No other disclosures were reported.

Funding/Support: This is the result of work supported in part by resources from the Veterans Health Administration and grant 1I01HX003605 (awarded to Drs Gupta and May) from the Department of Veterans Affairs (VA) Health Services Research and Development; grant 1IK2CX002648 from the VA Clinical Science Research and Development (awarded to Dr Seligman); grants 5R37CA222866 (awarded to Dr Gupta), 5R01CA271034 (awarded to Dr May), and R00CA267181 (awarded to Dr Demb) from the National Cancer Institute; grant 5R03AG082989 (awarded to Dr Seligman) from the National Institute on Aging; and funding from the Kaiser Permanente Delivery Science and Applied Research Physician Researcher Program (awarded to Dr Levin). Support for data from the VA and the Centers for Medicare & Medicaid Services was provided by the Department of Veterans Affairs, Office of Research and Development, VA Information Resource Center (project No. SDR 02-237 and No. 98-004).

Role of the Funder/Sponsor: The funders/sponsors had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Disclaimer: The content is solely the responsibility of the authors and does not represent the views of the Department of Veterans Affairs or the US government.

Data Sharing Statement: See Supplement 2.

REFERENCES

1. US Centers for Disease Control and Prevention. WONDER: underlying cause of death by single race 2018-2022. Accessed November 15, 2024. https://wonder.cdc.gov/wonder/help/ucd-expanded.html

2. Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. *CA Cancer J Clin*. 2023;73(1):17-48. doi:10.3322/caac.21763

3. Rex DK, Anderson JC, Butterly LF, et al. Quality indicators for colonoscopy. *Am J Gastroenterol*. 2024;119(9):1754-1780. doi:10.14309/ajg.0000000000002972

4. Gupta S, Lieberman D, Anderson JC, et al. Recommendations for follow-up after colonoscopy and polypectomy: a consensus update by the US Multi-Society Task Force on Colorectal Cancer. *Gastroenterology*. 2020;158(4):1131-1153.e5. doi:10.1053/j.gastro.2019.10.026

5. Causada-Calo N, Bishay K, Albashir S, Al Mazroui A, Armstrong D. Association between age and complications after outpatient colonoscopy. *JAMA Netw Open*. 2020;3(6):e208958. doi:10.1001/jamanetworkopen.2020.8958

6. Tran AH, Man Ngor EW, Wu BU. Surveillance colonoscopy in elderly patients: a retrospective cohort study. *JAMA Intern Med*. 2014;174(10):1675-1682. doi:10.1001/jamainternmed.2014.3746

7. Calderwood AH, Tosteson TD, Wang Q, Onega T, Walter LC. Association of life expectancy with surveillance colonoscopy findings and follow-up recommendations in older adults. *JAMA Intern Med*. 2023;183(5):426-434. doi:10.1001/jamainternmed.2023.0078

8. Gupta S, Earles A, Bustamante R, et al. Adenoma detection rate and clinical characteristics influence advanced neoplasia risk after colorectal polypectomy. *Clin Gastroenterol Hepatol*. 2023;21(7):1924-1936.e9. doi:10.1016/j.cgh.2022.10.003

9. Davidson KW, Barry MJ, Mangione CM, et al; US Preventive Services Task Force. Screening for colorectal cancer: US Preventive Services Task Force recommendation statement. *JAMA*. 2021;325(19):1965-1977. doi:10.1001/jama.2021.6238

10. Fine JP, Gray RJ. A proportional hazards model for the subdistribution of a competing risk. *J Am Stat Assoc*. 1999;94(446):496-509. doi:10.1080/01621459.1999.10474144

11. Gray RJ. A class of K-sample tests for comparing the cumulative incidence of a competing risk. *Ann Stat*. 1988;16:1141-1154. doi:10.1214/aos/1176350951

12. Kalbfleisch JD, Prentice RL. *The Statistical Analysis of Failure Time Data*. John Wiley and Sons; 1980.

13. Aalen O. Nonparametric estimation of partial transition probabilities in multiple decrement models. *Ann Stat*. 1978;6(3):534-545. doi:10.1214/aos/1176344198

14. Orkaby AR, Nussbaum L, Ho YL, et al. The burden of frailty among US veterans and its association with mortality, 2002-2012. *J Gerontol A Biol Sci Med Sci*. 2019;74(8):1257-1264. doi:10.1093/gerona/gly232

15. Cheng D, DuMontier C, Yildirim C, et al. Updating and validating the US Veterans Affairs Frailty Index: transitioning from *ICD-9* to *ICD-10*. *J Gerontol A Biol Sci Med Sci*. 2021;76(7):1318-1325. doi:10.1093/gerona/glab071

16. Click B, Pinsky PF, Hickey T, Doroudi M, Schoen RE. Association of colonoscopy adenoma findings with long-term colorectal cancer incidence. *JAMA*. 2018;319(19):2021-2031. doi:10.1001/jama.2018.5809

17. He X, Hang D, Wu K, et al. Long-term risk of colorectal cancer after removal of conventional adenomas and serrated polyps. *Gastroenterology*. 2020;158(4):852-861.e4. doi:10.1053/j.gastro.2019.06.039

18. Lee JK, Jensen CD, Levin TR, et al. Long-term risk of colorectal cancer and related death after adenoma removal in a large, community-based population. *Gastroenterology*. 2020;158(4):884-894.e5. doi:10.1053/j.gastro.2019.09.039

19. Wieszczy P, Kaminski MF, Franczyk R, et al. Colorectal cancer incidence and mortality after removal of adenomas during screening colonoscopies. *Gastroenterology*. 2020;158(4):875-883.e5. doi:10.1053/j.gastro.2019.09.011

20. Duvvuri A, Chandrasekar VT, Srinivasan S, et al. Risk of colorectal cancer and cancer related mortality after detection of low-risk or high-risk adenomas, compared with no adenoma, at index colonoscopy: a systematic review and meta-analysis. *Gastroenterology*. 2021;160(6):1986-1996.e3. doi:10.1053/j.gastro.2021.01.214

21. Atkin W, Wooldrage K, Brenner A, et al. Adenoma surveillance and colorectal cancer incidence: a retrospective, multicentre, cohort study. *Lancet Oncol*. 2017;18(6):823-834. doi:10.1016/S1470-2045(17)30187-0

22. Williams GJ, Hellerstedt ST, Scudder PN, Calderwood AH. Yield of surveillance colonoscopy in older adults with a history of polyps: a systematic review and meta-analysis. *Dig Dis Sci*. 2022;67(8):4059-4069. doi:10.1007/s10620-021-07198-6

23. Lee JK, Roy A, Jensen CD, et al. Surveillance colonoscopy findings in older adults with a history of colorectal adenomas. *JAMA Netw Open*. 2024;7(4):e244611-e244611. doi:10.1001/jamanetworkopen.2024.4611

24. Dalmat RR, Ziebell RA, Kamineni A, et al. Risk of colorectal cancer and colorectal cancer mortality beginning one year after a negative fecal occult blood test, among screen-eligible 76- to 85-year-olds. *Cancer Epidemiol Biomarkers Prev*. 2023;32(10):1382-1390. doi:10.1158/1055-9965.EPI-23-0265

25. Dalmat RR, Ziebell RA, Kamineni A, et al. Risk of colorectal cancer and colorectal cancer mortality beginning ten years after a negative colonoscopy, among screen-eligible adults 76 to 85 years old. *Cancer Epidemiol Biomarkers Prev*. 2023;32(1):37-45. doi:10.1158/1055-9965.EPI-22-0581

26. Wang L, Mannalithara A, Singh G, Ladabaum U. Low rates of gastrointestinal and non-gastrointestinal complications for screening or surveillance colonoscopies in a population-based study. *Gastroenterology*. 2018;154(3):540-555.e8. doi:10.1053/j.gastro.2017.10.006

27. Ko CW, Riffle S, Michaels L, et al. Serious complications within 30 days of screening and surveillance colonoscopy are uncommon. *Clin Gastroenterol Hepatol*. 2010;8(2):166-173. doi:10.1016/j.cgh.2009.10.007

28. Rutter CM, Johnson E, Miglioretti DL, Mandelson MT, Inadomi J, Buist DS. Adverse events after screening and follow-up colonoscopy. *Cancer Causes Control*. 2012;23(2):289-296. doi:10.1007/s10552-011-9878-5

29. Day LW, Kwon A, Inadomi JM, Walter LC, Somsouk M. Adverse events in older patients undergoing colonoscopy: a systematic review and meta-analysis. *Gastrointest Endosc*. 2011;74(4):885-896. doi:10.1016/j.gie.2011.06.023

30. Calderwood AH; Dartmouth-Hitchcock Medical Center. Colonoscopy vs Stool Testing for Older Adults With Colon Polyps (COOP). ClinicalTrials.gov identifier: NCT05612347. Accessed January 28, 2026. https://clinicaltrials.gov/study/NCT05612347